COMP 2711: Solutions to Final Exam
2024 Spring Semester

Problem 1. (10 points)

Find an integer N such that $2^n > 3n^2$ whenever n is greater than N. Prove that your result is correct using mathematical induction.

Answer: We want to find an integer N such that $2^n > 3n^2$ for all $n > N$. We choose $N = 7$. We will use induction to prove that $2^n > 3n^2$ whenever $n \geq 8$.

Base case: For $n = 8$, we have $2^8 > 3(8)^2$, because $2^8 = 256$ and $3(8)^2 = 3 \cdot 64 = 192$.

Induction step: For the inductive hypothesis, suppose that $2^k > 3k^2$ holds for some integer $k \geq 8$. We want to show that $2^{k+1} > 3(k+1)^2$.

Applying the inductive hypothesis, we get $2^{k+1} = 2 \cdot 2^k > 2 \cdot 3k^2$. Thus

$$2^{k+1} > 2(3k^2) = 3(2k^2) = 3(k^2 + k^2) \geq 3(k^2 + 3k),$$

where in the last step we have used the fact that $k \geq 3$ (recall that $k \geq 8$). Thus

$$2^{k+1} \geq 3(k^2 + 3k) = 3(k^2 + 2k + k) \geq 3(k^2 + 2k + 1),$$

since $k \geq 1$ (recall that $k \geq 8$). Thus $2^{k+1} > 3(k+1)^2$, which completes the inductive step. Therefore, we have shown that $N = 7$ satisfies the condition that $2^n > 3n^2$ for all $n > N$.

Grading strategy:

(a) Correct N worth 3 points. Any $N \geq 7$ is correct.

(b) Correct base case worth 3 points.

(c) Correct inductive step worth 4 points.

Problem 2. (10 points)

Box A contains 2 black and 6 white marbles, and box B contains 4 black and 2 white marbles. A box is selected at random (with equal probability for both boxes), and then a marble is drawn at random from the selected box.

(a) (5 points) What is the probability that the drawn marble is black?

(b) (5 points) Given that the drawn marble is white, what is the probability that it came from box A?

Answer:

(a) Let $P(\text{black})$ be the probability that the drawn marble is black. We have

$$P(\text{black}) = P(A) \cdot P(\text{black} \mid A) + P(B) \cdot P(\text{black} \mid B)$$

Since the boxes are chosen with equal probability, we have $P(A) = P(B) = \frac{1}{2}$. Also, $P(\text{black} \mid A) = \frac{2}{8}$ and $P(\text{black} \mid B) = \frac{4}{6}$. Therefore,

$$P(\text{black}) = \frac{1}{2} \cdot \frac{2}{8} + \frac{1}{2} \cdot \frac{4}{6} = \frac{1}{8} + \frac{2}{6} = \frac{11}{24}.$$

(b) Let $P(A \mid \text{white})$ be the probability that the marble came from Box A given that it is white. We use Bayes' theorem:

$$\begin{aligned} P(A \mid \text{white}) &= \frac{P(A) \cdot P(\text{white} \mid A)}{P(\text{white})} \\ &= \frac{P(A) \cdot P(\text{white} \mid A)}{P(A) \cdot P(\text{white} \mid A) + P(B) \cdot P(\text{white} \mid B)} \end{aligned}$$

We already know $P(A) = P(B) = \frac{1}{2}$, $P(\text{white} \mid A) = \frac{6}{8}$, and $P(\text{white} \mid B) = \frac{2}{6}$. Substituting these quantities, we obtain

$$P(A \mid \text{white}) = \frac{\frac{1}{2} \cdot \frac{6}{8}}{\frac{1}{2} \cdot \frac{6}{8} + \frac{1}{2} \cdot \frac{2}{6}} = \frac{\frac{3}{8}}{\frac{3}{8} + \frac{1}{6}} = \frac{9}{13}.$$

Therefore, the probability that the marble came from Box A given that it is white is 9/13.

Grading strategy:

(a) Maximum 5 points. Each calculation error -1, other kind of error get 0.

(b) Maximum 5 points. Each calculation error -1, other kind of error get 0.

Problem 3. (15 points)

(a) (10 points) Given two independent random variables X and Y, prove that $E(XY) = E(X)E(Y)$. Do *not* use any facts taught in class except standard definitions in the field of probability theory.

(b) (5 points) The *covariance* of two random variables X and Y, is defined to be the expected value of the random variable $(X - E(X))(Y - E(Y))$, and is denoted by $\mathrm{Cov}(X, Y)$. Prove that if X and Y are independent random variables, then the covariance $\mathrm{Cov}(X, Y)$ is equal to zero.

Answer: (a) Since X and Y are independent, we have:

$$P(X = x \text{ and } Y = y) = P(X = x)P(Y = y).$$

The expectation of XY is:

$$E(XY) = \sum_{x,y} xyP(X = x \text{ and } Y = y).$$

Using independence, we get:

$$E(XY) = \sum_{x,y} xyP(X = x)P(Y = y).$$

This can be factored into:

$$E(XY) = \left(\sum_{x} xP(X = x)\right)\left(\sum_{y} yP(Y = y)\right).$$

Thus,

$$E(XY) = E(X)E(Y).$$

(b) The covariance of X and Y is defined as

$$\mathrm{Cov}(X, Y) = E((X - E(X))(Y - E(Y))).$$

Expanding this, we get

$$\mathrm{Cov}(X, Y) = E(XY - Y \cdot E(X) - X \cdot E(Y) + E(X) \cdot E(Y)).$$

Using the principle of linearity of expectation, we obtain

$$\mathrm{Cov}(X, Y) = E(XY) - E(Y \cdot E(X)) - E(X \cdot E(Y)) + E(E(X) \cdot E(Y)),$$

which simplifies to

$$\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y) = E(XY) - E(X)E(Y).$$

From part (a), we know that for independent random variables X and Y, we have $E(XY) = E(X)E(Y)$. Therefore,

$$\mathrm{Cov}(X, Y) = 0.$$

Grading strategy:

(a) maximum 10 points. If only mention discrete or continuous variables, deduct 3 points. If use something not in the definition, deduct 5 points.

(b) maximum 5 points, any calculus error -1. Any other error would got 0.

Problem 4. (10 points)

You want to walk from the point $(0, 0)$ to the point $(8, 8)$ on a square grid, where each point on the grid has integer coordinates. In each step, you can move from a point (i, j) to either $(i+1, j)$ or $(i, j+1)$, which means you can move one unit in the positive x-direction or one unit in the positive y-direction, respectively. The walk will take exactly 16 unit steps in total. Additionally, your path must include the grid position $(2, 3)$ and you must avoid the grid position $(6, 5)$. How many different paths can you take from $(0, 0)$ to $(8, 8)$ while satisfying these conditions?

Answer: We say that a path is *monotonic* if at each step it either moves one unit in the positive x-direction or one unit in the positive y-direction. Let P be a path satisfying the requirements of the problem. Since it must include the point $(2, 3)$, we can decompose P into two monotonic paths P_1 and P_2 as follows:

$$\overbrace{(0,0) \to \cdots \to \underbrace{(2,3)}}^{P_1} \to \cdots \to \underbrace{(8,8)}_{P_2}.$$

Our answer will be $\#P_1 \cdot \#P_2$, where $\#P_1$ ($\#P_2$) is the number of possibilities for P_1 (P_2). It is clear that the requirement of P avoiding $(6, 5)$ only restricts the possibilities for P_2 but not P_1. Thus, $\#P_1$ is simply the number of all monotonic paths from $(0, 0)$ to $(2, 3)$, which is given by

$$\binom{2+3}{2} = \binom{5}{2}.$$

We can count $\#P_2$ as

$$\#P_2 = \#P_{2,all} - \#P_{2,bad}$$

where $\#P_{2,all}$ is the number of all monotonic paths from $(2, 3)$ to $(8, 8)$ and $\#P_{2,bad}$ is the number of 'bad' monotonic paths from $(2, 3)$ to $(8, 8)$ that go through $(6, 5)$.

Using the same reasoning for counting $\#P_1$, we get that:

$$\#P_{2,all} = \binom{(8-2)+(8-3)}{8-2} = \binom{11}{6}.$$

Similar to P, the paths counted by $\#P_{2,bad}$ have the form:

$$\overbrace{(2,3) \to \cdots \to \underbrace{(6,5)}}^{P_3} \to \cdots \to \underbrace{(8,8)}_{P_4},$$

where P_3 and P_4 are monotonic paths. The number of possibilities for P_3 is

$$\binom{(6-2)+(5-3)}{6-2} = \binom{6}{4},$$

and the number of possibilities for P_4 is

$$\binom{(8-6)+(8-5)}{8-6} = \binom{5}{2}.$$

Overall, we get that

$$\#P_{2,bad} = \binom{11}{6} - \binom{6}{4} \cdot \binom{5}{2},$$

which implies that our final answer is:

$$\binom{5}{2} \cdot \left[\binom{11}{6} - \binom{6}{4} \cdot \binom{5}{2} \right].$$

Grading strategy:

- 4 marks for showing, at any capacity, that the number of paths from a to b that go through c is $\#(a \rightsquigarrow c) \cdot \#(c \rightsquigarrow b)$.
- 4 marks for showing, at any capacity, that the number of paths from a to b that avoid c is $\#(a \rightsquigarrow b) - \#(a \rightsquigarrow c) \cdot \#(c \rightsquigarrow b)$.
- 2 points for putting the previous two points together to devise a correct solution.

Problem 5. (10 points)

You have a set of n biased coins. For each $m = 1, 2, \ldots, n$, the m-th coin has a probability of $\frac{1}{2m+1}$ of landing heads when tossed, and the outcomes of the tosses for each coin are independent of each other.

Show that the probability of obtaining an odd number of heads after tossing all the n coins is equal to $\frac{n}{2n+1}$.

Hint: One possible approach is to think in a recursive way.

Solution: Let P_n be the probability of obtaining an odd number of heads when tossing these n coins. In order to compute P_{n+1}, we need to consider two cases:

Case 1: The $(n+1)$-th coin lands head and we have an even number of heads in the first n coins.

Case 2: The $(n+1)$-th coin lands tail and we have an odd number of heads in the first n coins.

Thus:

$$P_{n+1} = \left(\frac{1}{2n+3}\right)(1 - P_n) + \left(1 - \frac{1}{2n+3}\right) P_n = \left(1 - \frac{2}{2n+3}\right) P_n + \frac{1}{2n+3}.$$

Multiplying both sides by $(2n+3)$ we obtain

$$(2n+3)P_{n+1} = (2n+1)P_n + 1.$$

Substituting $A_n = (2n+1)P_n$, we can express this as $A_{n+1} = A_n + 1$. We know that $A_0 = 0$ because $P_0 = 0$. This implies that $A_n = n$, which in turn implies that $P_n = \frac{n}{2n+1}$, as desired.

Grading strategy:

- 5 points for presenting the recurrence relation.
- 5 points for solving the recurrence relation.

Other solutions too might be correct.

Problem 6. (10 points)

Consider tossing three fair six-sided dice at the same time, where each dice shows a number between 1 and 6. What is the probability that the three numbers on the dice can be rearranged to form three consecutive integers?

Solution: To find the probability that the three numbers on the dice can be rearranged to form three consecutive integers, we need to determine the favorable cases and divide by the total number of possible outcomes. There are 4 cases where the three dice show consecutive numbers:

$\{1, 2, 3\}$

$\{2, 3, 4\}$

$\{3, 4, 5\}$

$\{4, 5, 6\}$

For each of these cases, there are $3! = 6$ possible permutations of the three numbers. Therefore, the total number of favorable outcomes is $4 \times 6 = 24$.

The total number of possible outcomes when rolling three fair six-sided dice is $6^3 = 216$ (each die has 6 possible outcomes, and there are 3 dice).

Thus, the probability of getting three consecutive numbers is:

$$\frac{24}{216} = \frac{1}{9}.$$

Grading strategy:

- 4 points for presenting all possibilities.
- 6 points for calculating the probability for each.

Problem 7. (10 points)

A binary string is a string consisting of 0s and 1s. A *run* is a maximal substring of consecutive 1s. For example, the binary string $\underline{11}0\underline{1}0\underline{111}0\underline{11}$ has 4 runs, which are underlined.

Given positive integers n, m, and k, determine the number of binary strings that simultaneously satisfy all of the following three conditions: the string contains exactly m 0s, the string contains exactly n 1s, and the string has exactly k runs of consecutive 1s.

Solution: We first lay out all the m 0s which are surrounded by boxes that will later hold 1s:

$$\underbrace{\square\, 0\, \square\, 0 \ldots 0\, \square\, 0\, \square}_{m \text{ 0s and } m+1 \text{ }\square\text{'s}}.$$

Note that we have boxes between 0s, as well as to the left of the leftmost 0 and to the right of the rightmost 0. We define a box to hold an arbitrary number of 1s. Observe the following: any string with k runs chooses exactly k out of the $m+1$ boxes and maps the k runs to the k boxes by putting at least one 1 in each such box while leaving the remaining $m+1-k$ boxes empty. Thus, we can choose the positions of runs with respect to the 0s in $\binom{m+1}{k}$ ways.

After choosing the k boxes that hold the k runs, it remains to distribute the n 1s among the k runs. We resolve this issue using the stars and bars kind of reasoning. By definition, each box must have at least one 1, and we want the sum of the number of 1s in the k boxes to be n. The number of ways to distribute the n 1s over the k runs is, therefore, the same as the number of permutations of $n-k$ stars and $k-1$ bars, which is given by

$$\binom{(n-k)+(k-1)}{k-1} = \binom{n-1}{k-1}.$$

Therefore, we conclude that the final answer is

$$\binom{m+1}{k} \cdot \binom{n-1}{k-1}.$$

Grading strategy:

- 2 points for approaching the problem as an arrangement between the 0s and runs/1s, even if this does not make concrete progress towards the answer.
- 4 points finding the number of ways to assign the runs to boxes.
- 4 for showing the number of ways of distributing 1s among the runs.

Problem 8. (10 points) Let n be a positive integer such that $n = p^2q^4r^2$, where p, q, and r are three distinct prime numbers. Prove that the number of integers from 1 to n that are relatively prime with n is given by the expression:

$$n\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right)\left(1-\frac{1}{r}\right).$$

Solution: To find the number of integers from 1 to n that are relatively prime to n, we need to count the number of integers from 1 to n that are not divisible by p, q, or r. Let $\phi(n)$ denote this quantity.

Let $S = \{1, 2, \ldots, n\}$. Define:

A_p as the set of integers in S divisible by p,
A_q as the set of integers in S divisible by q,
A_r as the set of integers in S divisible by r.

Using the principle of inclusion-exclusion, the number of integers in S that are divisible by p, q, or r is given by:

$$|A_p \cup A_q \cup A_r| = |A_p| + |A_q| + |A_r| - |A_p \cap A_q| - |A_p \cap A_r| - |A_q \cap A_r| + |A_p \cap A_q \cap A_r|.$$

We need to determine the size of each of these sets:

$$\begin{aligned}
|A_p| &= \frac{n}{p}, \\
|A_q| &= \frac{n}{q}, \\
|A_r| &= \frac{n}{r}, \\
|A_p \cap A_q| &= \frac{n}{pq}, \\
|A_p \cap A_r| &= \frac{n}{pr}, \\
|A_q \cap A_r| &= \frac{n}{qr}, \\
|A_p \cap A_q \cap A_r| &= \frac{n}{pqr}.
\end{aligned}$$

Substituting these values, we get:

$$|A_p \cup A_q \cup A_r| = \frac{n}{p} + \frac{n}{q} + \frac{n}{r} - \frac{n}{pq} - \frac{n}{pr} - \frac{n}{qr} + \frac{n}{pqr}.$$

The number of integers in S that are relatively prime to n is:

$$\phi(n) = n - |A_p \cup A_q \cup A_r|.$$

Substituting the expression for $|A_p \cup A_q \cup A_r|$, and simplifying, we get:

$$\begin{aligned}
\phi(n) &= n - \left(\frac{n}{p} + \frac{n}{q} + \frac{n}{r} - \frac{n}{pq} - \frac{n}{pr} - \frac{n}{qr} + \frac{n}{pqr}\right) \\
&= n\left(1 - \left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r} - \frac{1}{pq} - \frac{1}{pr} - \frac{1}{qr} + \frac{1}{pqr}\right)\right) \\
&= n\left(1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r} + \frac{1}{pq} + \frac{1}{pr} + \frac{1}{qr} - \frac{1}{pqr}\right) \\
&= n\left(1 - \frac{1}{p}\right)\left(1 - \frac{1}{q}\right)\left(1 - \frac{1}{r}\right).
\end{aligned}$$

Grading strategy:

- 5 points for defining the inclusion and exclusion sets.
- 3 points for calculating the size of each sets.
- 2 points for final calculations.

Problem 9. (15 points)

(a) (5 points) Let p be a prime number and k be an integer such that $1 \leq k \leq p-1$. Prove that the binomial coefficient $\binom{p}{k}$ is divisible by p.

(b) (10 points) Let p be a prime number. Use mathematical induction to prove that for any positive integer a, the number $a^p - a$ is divisible by p. In other words, you need to prove Fermat's Little Theorem using induction on the integer a.

Note: If you give the number-theoretic proof of Fermat's Little Theorem presented in class, you will receive 0 marks.

Hint: In your proof of part (b), you may find the claim given in part (a) useful. Note that in part (b), you are permitted to use this claim from part (a), even if you were unable to prove it.

Solution: (a) Consider the binomial coefficient:

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

The numerator $p!$ can be written as $p \cdot (p-1) \cdots 2 \cdot 1$, which is clearly divisible by p. The denominator is the product of two factorials, $k!$ and $(p-k)!$. Since $1 \leq k \leq p-1$, both k and $p-k$ are integers strictly less than p. Thus, the denominator $k!(p-k)!$ is the product of integers all of which are non-zero and strictly less than p. This means that neither $k!$ nor $(p-k)!$ is divisible by p.

Since $\binom{p}{k}$ is an integer, and it is expressed as a fraction where the numerator is divisible by p but the denominator is not, we can conclude that $\binom{p}{k}$ itself must be divisible by p.

(b) The base case is when $a = 1$, in which case we have $a^p - a = 1^p - 1 = 0$, which is divisible by p, and hence the base case holds. Now supposing that the claim holds for an arbitrary a, i.e., $p \mid a^p - a$, we wish to show that it holds for $a+1$, i.e., $p \mid (a+1)^p - (a+1)$. Expanding $(a+1)^p - (a+1)$ using the binomial formula, we get

$$
\begin{aligned}
(a+1)^p - (a+1) &= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i \cdot 1^{p-i} \right) - (a+1) \\
&= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i \right) - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i \right) + \binom{p}{0} \cdot a^0 + \binom{p}{p} \cdot a^p - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i \right) + \cancel{1} + a^p - (a + \cancel{1}) \\
&= \left(\sum_{i=1}^{p-1} \underbrace{\binom{p}{i}}_{\substack{\text{divisible by } p \\ \text{from part (a)}}} \cdot a^i \right) + \underbrace{a^p - a}_{\text{divisible by } p \text{ by IH}} .
\end{aligned}
$$

Thus, $(a+1)^p - (a+1)$ is equal to the sum of terms each of which is divisible by p, and therefore we have $p \mid (a+1)^p - (a+1)$, which completes our inductive proof.

Grading strategy:

(a) 3 points for showing that the numerator is divisible by p, and 2 points for showing that the denominator is not divisible by p.

(b) 2 marks for the base case; 3 marks for binomial expansion; and 5 marks for carrying out the calculations carefully. 3 marks are deducted if a solution derives the answer by assuming that all binomial coefficients (including $\binom{p}{0}$ and $\binom{p}{p}$) are divisible by p.